

DAP ES9038Q2M Clocking & I²S Interface Rev7

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Attribution

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Part Identification

This document defines the digital audio link and clocking between the ESP32-S3 (source) and the ES9038Q2M (32-bit stereo DAC, ESS). It covers which device is timing master, the pin mapping, and the clock sources.

- **Source:** ESP32-S3. It reads files and outputs PCM over I²S.
- **DAC:** ES9038Q2M. It converts PCM to analog and feeds the OPA1622 output stage.
- **Reference:** ES9038Q2M datasheet v1.4, pp. 4, 6, 7, 10.

Design Requirements

- **Input format:** PCM only, over I²S (2-channel serial).
- **Sample-rate families:** both must be supported.
 - 44.1 kHz family: 44.1, 88.2, 176.4 kHz, and so on.
 - 48 kHz family: 48, 96, 192 kHz, and so on.
- **Timing roles:** the ES9038Q2M is I²S master (master_mode = 1, Reg 10). The ESP32-S3 is I²S slave.
- **DPLL:** off, synchronous mode (128fs_mode = 1, Reg 10).
- **Clocking:** two external tri-state oscillators, one per family, with outputs tied together at XI. Only one is enabled at a time.
- **XI frequency range:** $XI \leq 100$ MHz, and $XI/2^{\text{clk_gear}} = 128 \cdot \text{FSR}$ exactly. The exact value is not critical; the selection criterion is lowest noise.

Design Constraints

- Low noise
- Small footprint
- No DSD input (native DSD and DoP are both excluded)
- No S/PDIF input
- ESP32-S3 platform; no bare-metal requirement
- DAC I²C: $SCL < MCLK/20$ (p. 14). At the lowest MCLK (5.6448 MHz) this is ≤ 282 kHz, so run I²C at 100 kHz standard mode.
- No clock switching during playback. Mute the DAC or hold RESETB low while switching (p. 11).

Pin Configuration

With the DAC as master, it drives BCLK and LRCLK out to the ESP32-S3, and only data flows into the DAC. The ES9038Q2M has no MCLK input pin; its internal MCLK is always derived from XI (datasheet p. 6).

Pin	Name	Function (master, PCM)	Driven by
20	DATA_CLK	Bit clock (BCLK) output	ES9038Q2M
19	DATA1	Frame clock (LRCLK/WS) output	ES9038Q2M
18	DATA2	Serial PCM data (SIN) input	ESP32-S3
29	XI	System clock input, 100 MHz max	External oscillators (2, one per family)
28	XOUT	Crystal output; unused when XI is driven by an oscillator	—
12, 27	VCCA	+3.3 V oscillator supply	Clean 3.3 V rail
16, 17	GPIO2, GPIO1	Unused for audio in this mode; either can output buffered MCLK (Reg 8)	—
23, 24, 25	SCL, SDA, ADDR	I ² C control and address select	ESP32-S3
21	RESETB	Reset/power-down, active low	ESP32-S3 GPIO

Data pins, DATA_CLK, GPIO1–2, SCL, SDA, ADDR, and RESETB are 5 V tolerant only when DVCC = 3.3 V (p. 5).

External Components

Values assume a top playback rate of 176.4/192 kHz. For 352.8/384 kHz, both oscillator frequencies double.

Ref	Part	Value	Notes
X1	ECS-2520MVLC-225.792-CN-TR (ECS, XO, 2.5 × 2.0 mm)	22.5792 MHz	44.1 kHz family; –159.8 dBc/Hz @ 100 kHz (typ. plot)
X2	ASTX-H11-24.576MHZ-T (Abracon, TCXO)	24.576 MHz	48 kHz family; –158 dBc/Hz @ 100 kHz

Ref	Part	Value	Notes
R_S1, R_S2	Series resistor, one per oscillator output	22 Ω	Placed at each oscillator output pin; outputs join at XI after the resistors
R_EN1, R_EN2	Pull-down resistor, one per EN line	10 k Ω	EN to GND; both oscillators disabled at power-up
R_RST	Pull-down resistor on RESETB	TBD	Holds the DAC in reset until firmware releases it

- Each OE pin is driven by its own ESP32-S3 GPIO. Exactly one oscillator is enabled at a time.
- On both parts, pulling pin 1 low puts the output in high-impedance, and pulling it high or leaving it unconnected enables it.
 - X1: $\leq 0.3 \cdot V_{dd}$ gives Hi-Z; $\geq 0.7 \cdot V_{dd}$ or NC gives active output.
 - X2: $< 0.25 \cdot V_{dd}$ gives Hi-Z; $\geq 0.75 \cdot V_{dd}$ or open gives active output.
- XOUT (pin 28) is left unconnected.

Power-up state: both EN lines and RESETB are pulled low. Both oscillator outputs are Hi-Z, XI is idle, and the DAC stays in reset until the firmware enables one oscillator and releases RESETB.

Clock Sources

In the DAC's default asynchronous mode, the clock that times the analog conversion is **XI**, not the ESP32-S3's BCLK/LRCLK. The DPLL measures the incoming frame rate against MCLK, and the Jitter Eliminator strips the input timing error (p. 10). Both chips can generate clocks, but only the XI source directly sets output jitter.

ESP32-S3 I²S clock sources ([ESP-IDF Clock Tree](#), [ESP32-S3](#)):

Source	Frequency	Origin
PLL_F160M (default)	160 MHz fixed	PLL multiplying the 40 MHz crystal
PLL_D2	240 MHz (160 MHz if PLL = 320 MHz)	Same PLL, divided by 2

Source	Frequency	Origin
XTAL	40 MHz	External crystal, no PLL
EXTERNAL	Set by the external clock	Clock supplied from off-chip

- The internal ~17.5 MHz RC oscillator (RC_FAST) is not an I²S source. It only feeds low-power and RTC functions.
- The ESP32-S3 has no audio PLL (APLL), unlike the original ESP32.
- A PLL does not "auto-adjust" toward a better clock. It locks a fast oscillator to a reference and multiplies that reference. Its output is at best as clean as the reference within its loop bandwidth, plus its own VCO noise outside that bandwidth.

Generic jitter ranking, lowest to highest:

1. A dedicated low-phase-noise crystal oscillator (XO) at an exact audio multiple, with no PLL and an integer divider only.
2. A crystal on the chip's built-in oscillator (XI/XOUT).
3. A PLL output with an integer divider.
4. A PLL output with a fractional divider, when the target frequency is not an integer division of the PLL.
5. An RC oscillator.

Equations and Results

The DAC's internal MCLK comes from XI through the `clk_gear` divider (Reg 0, p. 15):

$$\text{MCLK} = \frac{\text{XI}}{2^{\text{clk_gear}}}, \quad \text{clk_gear} \in \{0, 1, 2, 3\}$$

The bit clock is set by the serial word length (Reg 1, p. 6):

$$f_{\text{BCLK}} = 2 \cdot \text{serial_length} \cdot \text{FSR}$$

Minimum MCLK depends on the mode (Analog Performance, p. 52):

$$\text{MCLK}_{\text{min,async}} = 192 \cdot \text{FSR} \quad \text{qqquad} \quad \text{MCLK}_{\text{min,sync}} = 128 \cdot \text{FSR}$$

Mode	Ratio	44.1 kHz	48 kHz
Synchronous (DPLL bypassed)	128·FSR (exact)	5.6448 MHz	6.144 MHz
Asynchronous (default, DPLL on)	≥ 192·FSR	≥ 8.4672 MHz	≥ 9.216 MHz

- The 128·FSR values (5.6448 and 6.144 MHz) apply only to synchronous mode. In the default asynchronous mode, MCLK just has to clear 192·FSR at the highest rate you play.
- Maximum FSR is 384 kHz in asynchronous mode and 768 kHz in synchronous mode (p. 52).

clk_gear only divides, so XI must equal the highest MCLK in each family:

Oscillator (XI)	clk_gear ÷1	clk_gear ÷2	clk_gear ÷4
22.5792 MHz	176.4 kHz	88.2 kHz	44.1 kHz
24.576 MHz	192 kHz	96 kHz	48 kHz

The firmware reads each track’s sample rate, enables the matching oscillator, and writes clk_gear over I²C.

Open Problems

- ☐ **Maximum playback rate.** Pick the highest FSR you will support. It sets the minimum MCLK and whether you can stay in asynchronous mode (384 kHz cap).

Rev Log

Rev	Date	Change
7	2026-09-23	R_S1/R_S2 set to 22 Ω, placed at each oscillator output. Series resistor open problem closed.
6	2026-09-23	Added 10 kΩ pull-downs on both oscillator EN lines and a pull-down on RESETB. Power-up state recorded.

Rev	Date	Change
5	2026-09-23	Oscillator parts selected: X1 ECS-2520MVLC-225.792-CN-TR, X2 ASTX-H11-24.576MHZ-T. Pin 1 enable behavior recorded. Oscillator part selection open problem closed.
4	2026-09-23	Oscillator values (22.5792 / 24.576 MHz) and clk_gear mapping recorded. Tri-state oscillators tied at XI. Added External Components, I ² C speed limit, and the clock-switching rule.
3	2026-09-23	Switched to DAC master / ESP32-S3 slave with the DPLL off (synchronous mode). Pin table updated. Resolved and moot open problems removed.
2	2026-09-23	ESP32-S3 master confirmed. XI frequency range recorded; oscillators to be selected for lowest noise.
1	2026-09-23	Clocking set to two external oscillators, one per sample-rate family. XI source-type open problem closed.
0	2026-09-23	Initial release. PCM-only interface, ESP32-S3 master / DAC slave, pin mapping, clock source survey, MCLK equations, and open clocking decisions.

Sources

- ES9038Q2M Datasheet v1.4 (ESS, Jan 2021), pp. 4–6, 10, 15, 24, 52 (project file)
- [ESP-IDF Clock Tree, ESP32-S3](#)